6.Perform basic Image Handling and processing operations on the image is to read an image in python and Convert an Image to erode using Erode function.

Aim

To write a Python program to read an image and perform image erosion using the cv2.erode() function in OpenCV.

Procedure

1. Import the OpenCV library.
2. Read the input image.
3. Convert the image into grayscale.
4. Apply binary thresholding to create a binary image.
5. Create a kernel using cv2.getStructuringElement().
6. Apply erosion using cv2.erode().
7. Display the original binary image and eroded image.

Code

```
import cv2
import numpy as np

img = cv2.imread("sample.jpg", 0)

if img is None:
    print("Image not found")
else:
    _, binary = cv2.threshold(
        img, 127, 255, cv2.THRESH_BINARY
    )

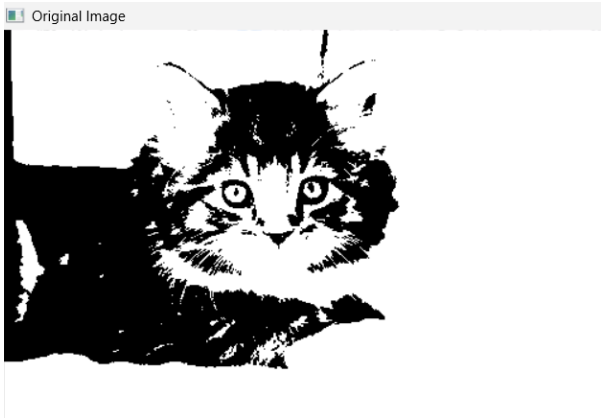
    kernel = np.ones((5, 5), np.uint8)
    eroded = cv2.erode(binary, kernel, iterations=1)

    cv2.imshow("Original Image", binary)
```

```
cv2.imshow("Eroded Image", eroded)


cv2.waitKey(0)
cv2.destroyAllWindows()
```

SAMPLE INPUT:



SAMPLE OUTPUT:



Result

Thus, image erosion was successfully performed using the cv2.erode() function. The white regions in the binary image became smaller.